The quotient questions for f-algebras

A literature resolution and elementary verification of Questions 8.3 in arXiv:2005.00902

Automated mathematical discovery run `fa_banach_001`, lane 9

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Disposition: literature already answered. All three questions have affirmative answers. The later paper Muñoz–Lahoz–Tradacete, arXiv:2511.13299, explicitly records that f-algebras form an equational class, citing Birkhoff's *Lattice Theory*, Theorem VII.8, and notes that free f-algebras have been known since 1956. The proof below independently verifies the claim and displays two explicit defining identities, but is not asserted to be new.

Source question and transcription

Let A be an f-algebra, let B be a vector lattice algebra, and let $\varphi : A \to B$ be a surjective vector lattice algebra homomorphism. Questions 8.3 ask:

1. Must B be an f-algebra?
2. Is the answer still affirmative when A is unital?
3. Is the answer still affirmative when A is unital with positive identity?

The source observes that affirmative answers make the corresponding three classes equational and yield free objects over nonempty sets.

Definitions

A vector lattice algebra is an associative real algebra which is also a vector lattice and whose positive cone is closed under multiplication. It is an f-algebra when, for $x, y \in A$ and $z \in A_+$,

$$x \wedge y = 0 \quad \Longrightarrow \quad (xz) \wedge y = (zx) \wedge y = 0.$$

A vector lattice algebra homomorphism preserves the linear, lattice, and algebra operations. The kernel of a surjective such homomorphism is a two-sided algebra ideal and an order ideal.

Idea / proof intuition

There are two short routes. First, replace arbitrary positive elements x, y by the disjoint remainders $x - x \wedge y$ and $y - x \wedge y$. This turns the conditional f-algebra axiom into two unconditional identities. Second, in a quotient write $x = u + x_0$, $y = u + y_0$, where $u = x \wedge y$ lies in the kernel and $x_0 \perp y_0$. After multiplying, the only term not visibly controlled by the kernel vanishes by the f-algebra property.

Original source image

A more indirect way to prove that they *are* equational classes would be to try to use Birkhoff's theorem (see [7, Theorem 4.41]), which shows that the equational classes are precisely the varieties of abstract algebras. We recall that a variety of abstract algebras is a class of abstract algebras, all of the same type, that is closed under taking abstract subalgebras, abstract product algebras, and abstract homomorphic images. It is clear that each of the classes of f-algebras, of unital f-algebras, and of unital f-algebras with a positive identity element are closed under taking (unital) vector lattice subalgebras and taking vector lattice algebraic products. It seems to be open, however, whether any of these three classes is closed under the taking of abstract algebra homomorphic images, i.e., under (unital) vector lattice algebra homomorphic images. Hence we have the following questions.

Questions 8.3. Let A be an f-algebra, let B be a vector lattice algebra, and let $\varphi : A \to B$ be a surjective vector lattice algebra homomorphism.

(1) Is B always an f-algebra? If so, then the f-algebras form an equational class and free f-algebras over non-empty sets exist.
(2) Is B always an f-algebra (automatically unital) when A is unital? If so, then the unital f-algebras form an equational class and free unital f-algebras over non-empty sets exist.
(3) Is B always an f-algebra (automatically unital with a positive identity element) when A is unital with a positive identity element? If so, then the unital f-algebras with a positive identity element form an equational class and free unital f-algebras with a positive identity element over non-empty sets exist.

We have no answers. Related to all three questions, we can only mention that it is known that the answers are affirmative, *provided* that we also know that both A and B are Archimedean; this follows from [11, Proposition 3.2]. Related to the second and third question, we can only mention that it is known that an Archimedean vector lattice algebra with an identity element is an f-algebra precisely when all squares are positive; see [21, Corollary 1]. This shows once more that the answers to the second and third questions are affirmative, *provided* that we also know that both A and B are Archimedean. Unfortunately, this is not what we need, and to the best of our knowledge these three issues are all open at the time of writing.

Even though the Archimedean f-algebras, the Archimedean unital f-algebras, and the Archimedean unital f-algebras with a positive identity are not an equational class, and the f-algebras, the unital f-algebras, or the unital f-algebras with a positive identity element might also not be, this does not preclude the possibility that one or more of these six classes still has free objects over non-empty sets. After all, the Archimedean vector lattices do not form an equational class because there exist quotients of such lattices that are no longer Archimedean, but free Archimedean vector lattices over non-empty sets still exist. The reason is simply that the general free vector lattice over a non-empty set, which has

Source: de Jeu, arXiv:2005.00902, PDF p. 35, Questions 8.3. This is a direct crop from `source_paper.pdf`.

Later literature answer

The later paper *The free Banach f-algebra generated by a Banach space* states that the existence of free f-algebras is long known and follows from the fact that f-algebras form an equational class. It cites Keimel's 1995 survey, Section 5.2, and Birkhoff's 1961 *Lattice Theory*, Theorem VII.8. Its introduction also attributes the foundational theory to Birkhoff–Pierce (1956).

CHAPTER 2

Free Archimedean f-algebras

This chapter is devoted to the construction and study of the following object.

Definition 2.1. Let S be a set. The *free (Archimedean) f-algebra generated by S* is an (Archimedean) f-algebra $\mathrm{FA}f\mathrm{A}(S)$ together with a map $\delta\colon S \to \mathrm{FA}f\mathrm{A}(S)$ such that, for every (Archimedean) f-algebra A and every map $T\colon S \to A$, there exists a unique lattice-algebra homomorphism $\hat{T}\colon \mathrm{FA}f\mathrm{A}(S) \to A$ satisfying $\hat{T} \circ \delta = T$.

The existence of free f-algebras has been long known in the field of universal algebras [**Kei95**, Section 5.2]: it follows from the fact that f-algebras form an equationable class [**Bir61**, Theorem VII.8]. Archimedean f-algebras, however, do not form an equationable class [**dJ21**, Section 8]. Henriksen and Isbell showed in [**HI62**] that every LLA expression (that is, every expression involving finitely many variables, linear and lattice operations, and a product) that vanishes on the reals, also vanishes on every Archimedean f-algebra. It then follows from a standard argument due to Birkhoff [**Bir35**] that the free Archimedean f-algebra exists.

In Section 2.1 we revisit the result of Henriksen and Isbell through a completely new lens. Their proof and formulation depended heavily on universal algebra language and techniques that may be unfamiliar to the modern analyst. Here we shall use a more operator-centric approach together with a representation theorem due to Henriksen and Johnson [**HJ62**].

Admittedly, this may not seem very original. Yet this approach will be used in

Source: Muñoz–Lahoz–Tradacete, arXiv:2511.13299, PDF p. 23 (printed p. 11). This is a direct crop from the supporting PDF stored with this packet.

Elementary verification

For a, b in a vector lattice put

$$r(a,b) = a^+ - (a^+ \wedge b^+), \qquad s(a,b) = b^+ - (a^+ \wedge b^+).$$

Then $r(a,b), s(a,b) \geq 0$ and $r(a,b) \wedge s(a,b) = 0$.

Theorem 1 (Explicit equations and quotient stability)**.** *Among vector lattice algebras, the f-algebras are exactly those satisfying the two identities*

$$\big(r(a,b)c^+\big) \wedge s(a,b) = 0,$$
$$\big(c^+ r(a,b)\big) \wedge s(a,b) = 0$$

for all a, b, c. Consequently every surjective vector lattice algebra homomorphic image of an f-algebra is an f-algebra. The same holds in the unital and positive-unital signatures.

Proof. If A is an f-algebra, the displayed identities follow because $r(a,b)$ and $s(a,b)$ are positive and disjoint, while $c^+ \geq 0$. Conversely, suppose the identities hold and $x \wedge y = 0$, $z \geq 0$. Then $x, y \geq 0$, and substituting $(a,b,c) = (x,y,z)$ gives $r = x$, $s = y$, hence $(xz) \wedge y = (zx) \wedge y = 0$. Thus the identities are equivalent to the f-algebra axiom.

For a direct quotient check, let $q : A \to B$ be surjective and $I = \ker q$. Given $\bar{x}, \bar{y}, \bar{z} \geq 0$ in B with $\bar{x} \wedge \bar{y} = 0$, choose positive lifts x, y, z. Set $u = x \wedge y \in I$, $x_0 = x - u$, and $y_0 = y - u$. Then $x_0 \wedge y_0 = 0$, so $(x_0 z) \wedge y_0 = 0$. For positive elements of any vector lattice,

$$(a+b) \wedge (c+d) \leq (a \wedge c) + (a \wedge d) + (b \wedge c) + (b \wedge d),$$

which follows twice from the Riesz decomposition property. Therefore

$$\begin{aligned}(xz) \wedge y &= (uz + x_0 z) \wedge (u + y_0) \\ &\leq (uz \wedge u) + (uz \wedge y_0) + (x_0 z \wedge u) + (x_0 z \wedge y_0) \\ &\leq 2u + uz \in I.\end{aligned}$$

As I is an order ideal, $(xz) \wedge y \in I$, and hence $(\bar{x}\bar{z}) \wedge \bar{y} = 0$. Replacing right multiplication by left multiplication gives $(\bar{z}\bar{x}) \wedge \bar{y} = 0$. Thus B is an f-algebra. If A is unital, surjectivity sends its identity to the identity of B; positivity of the identity is likewise preserved. $\square$

Consequences, verification, and scope

- Questions 8.3(1)–(3) all have affirmative answers.
- The two displayed identities give an explicit equational axiomatization after adjoining the standard vector-lattice-algebra identities. The unital and positive-unital subclasses add the usual unit identities.
- Birkhoff's variety theorem therefore yields free f-algebras, free unital f-algebras, and free positive-unital f-algebras over nonempty sets.
- The proof is algebraic: it uses neither Archimedeanness nor a norm, and it treats noncommutative multiplication by retaining both left and right identities.

Verification. The positivity of lifts follows from surjectivity and $q(a^+) = q(a)^+$. The kernel is both an order ideal and a two-sided algebra ideal. The four-term meet inequality is valid in arbitrary (including non-Archimedean) vector lattices by Riesz decomposition. These checks remove the three main hidden failure modes in the quotient argument.

Novelty and limitations. This packet is a literature-resolution record, not a novelty claim. The later arXiv paper explicitly identifies the result as classical and gives the primary classical citations. No attempt was made here to settle the distinct Questions 8.4 asking whether particular free *vector lattice algebras* are themselves f-algebras or Archimedean.

Human review. Pending. Recommended check: compare the signature used in Birkhoff, Theorem VII.8, with the associative real vector lattice algebra signature of the source. The self-contained proof above independently covers exactly the source signature.

References

[1] M. de Jeu, *Free vector lattices and free vector lattice algebras*, arXiv:2005.00902 (2020/2021), Section 8.
[2] D. Muñoz–Lahoz and P. Tradacete, *The free Banach f-algebra generated by a Banach space*, arXiv:2511.13299 (2025/2026), Chapter 2.
[3] G. Birkhoff, *Lattice Theory*, revised ed., AMS Colloquium Publications XXV, 1961, Theorem VII.8.
[4] G. Birkhoff and R. S. Pierce, *Lattice-ordered rings*, An. Acad. Brasil. Ci. 28 (1956), 41–69.
[5] K. Keimel, *Some trends in lattice-ordered groups and rings*, 1995, Section 5.2.